Intensity Transformation Techniques

How pixel brightness gets reshaped — from photo negatives to contrast fixes

Point / intensity transformations in image processing

Five ways to bend the brightness curve

Every technique below takes an input intensity (0–255) and maps it to a new output intensity.

Negative

Flips dark ↔ light completely

Log transform

Expands shadow detail

Gamma / power-law

Tunable brighten or darken

Contrast stretch

Expands a narrow gray range

Histogram equalization

Auto-balances using the image itself

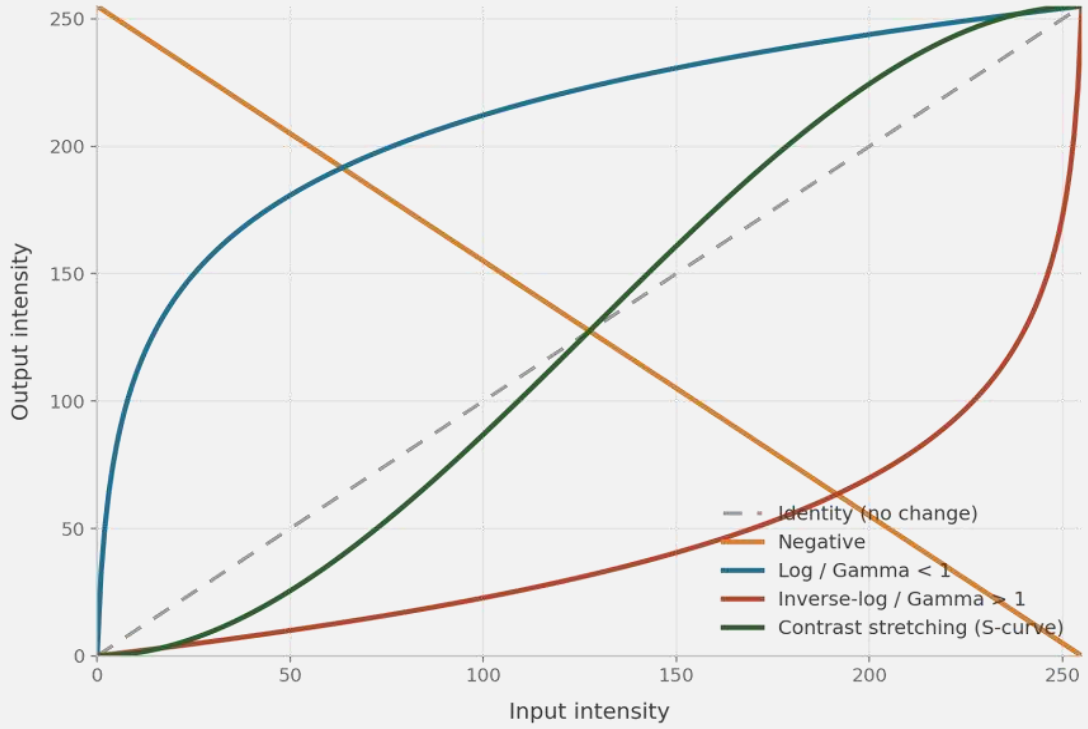


Image negative

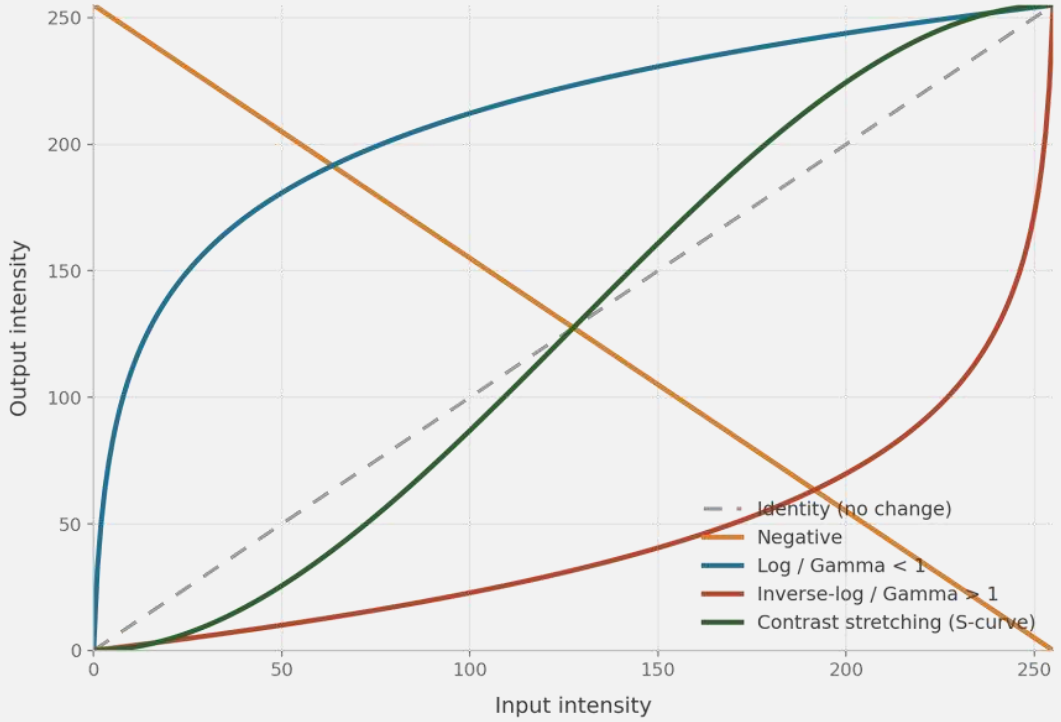
The photo film

What it does: reverses brightness completely — dark areas become bright, and bright areas become dark.

Analogy: an old-school film negative, or the “Invert colors” setting on a phone.

Best use: finding hidden detail in dark regions, like reading a faint X-ray.

```
output = 255 - input
```



log

Log transformation

The flashlight in a dark room

What it does

Expands dark pixel values while compressing bright ones — pulls hidden detail out of shadows without blowing out highlights.

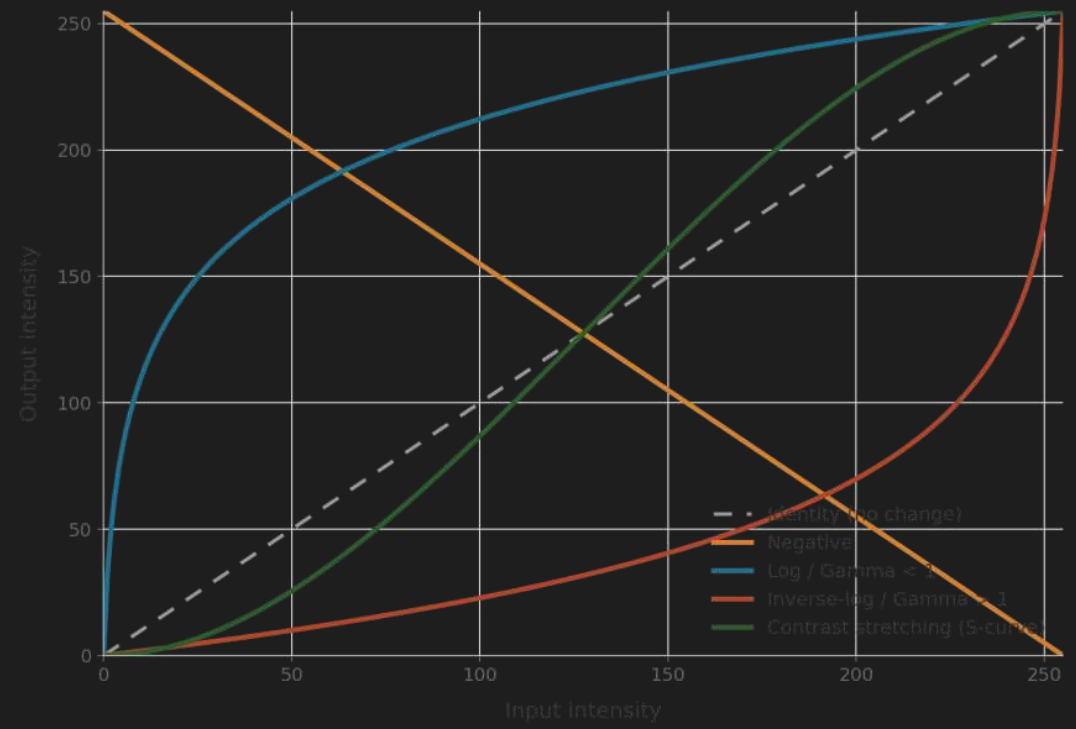
Analogy

Turning on a flashlight in a pitch-black room: dark details suddenly appear, but shining it on an already-bright wall changes little.

Best use

Displaying images with a huge intensity range, like Fourier spectra or radar imagery.

$$output = c \cdot \log(1 + input)$$



γ Gamma / power-law transform

The screen brightness slider

What it does: shifts midtones using a value γ (gamma).

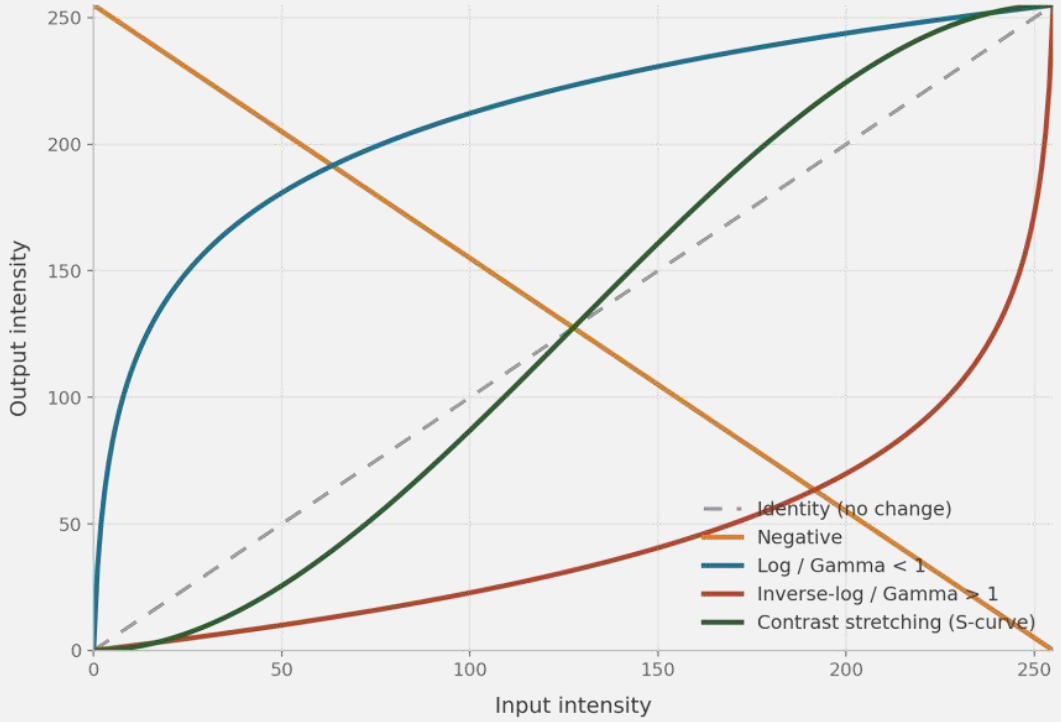
γ < 1 Brightens the image — stretches dark tones (behaves like the log curve).

γ > 1 Darkens the image — stretches bright tones (behaves like inverse-log).

Analogy: tweaking the brightness/contrast slider on a TV until the picture looks natural.

Best use: calibrating device displays so images don't look washed out or too dark.

```
output = 255 · (input / 255)^γ
```



Contrast stretching

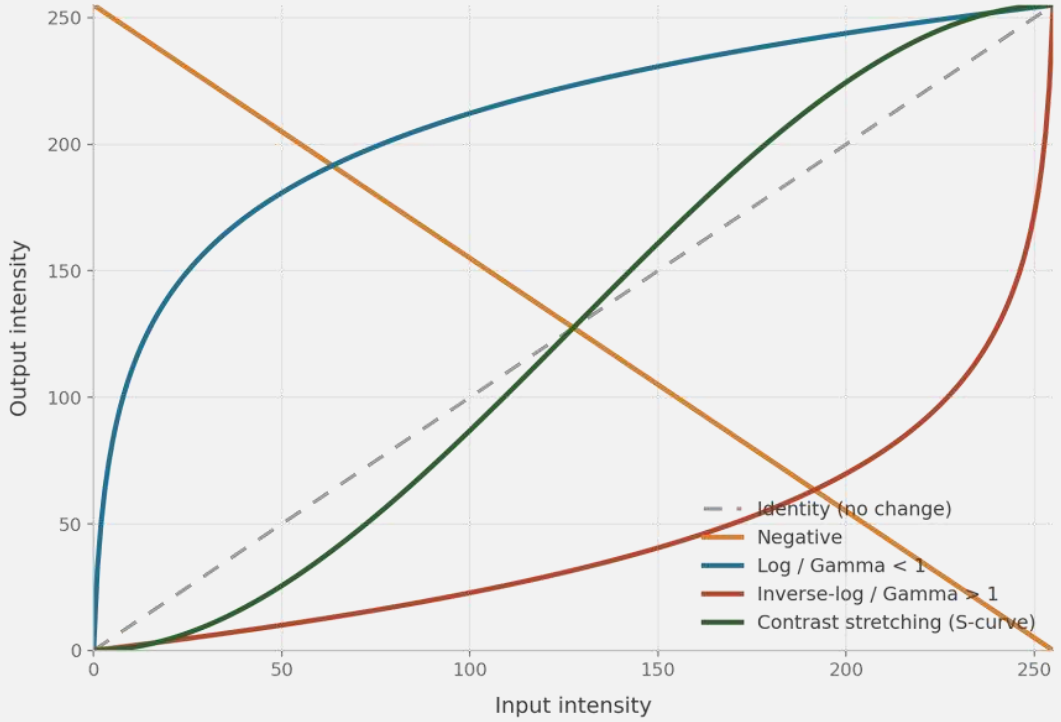
The elastic band

What it does: a piecewise-linear transform that takes a narrow range of grays and stretches it across the full 0–255 spectrum.

Analogy: pulling an elastic band — a crowded, blurry gray image gets its darks pulled lower and whites pushed higher until everything pops.

Best use: fixing poorly lit, washed-out, or low-contrast photos.

Note: the curve is a clean S-shape only when the stretch is centered on the midtones — shift the window toward shadows or highlights and it becomes asymmetric.



Histogram equalization

The auto-balance

What it does: instead of a fixed formula, it looks at the actual distribution of pixel values in this specific image and redistributes them so all intensity levels are used as evenly as possible.

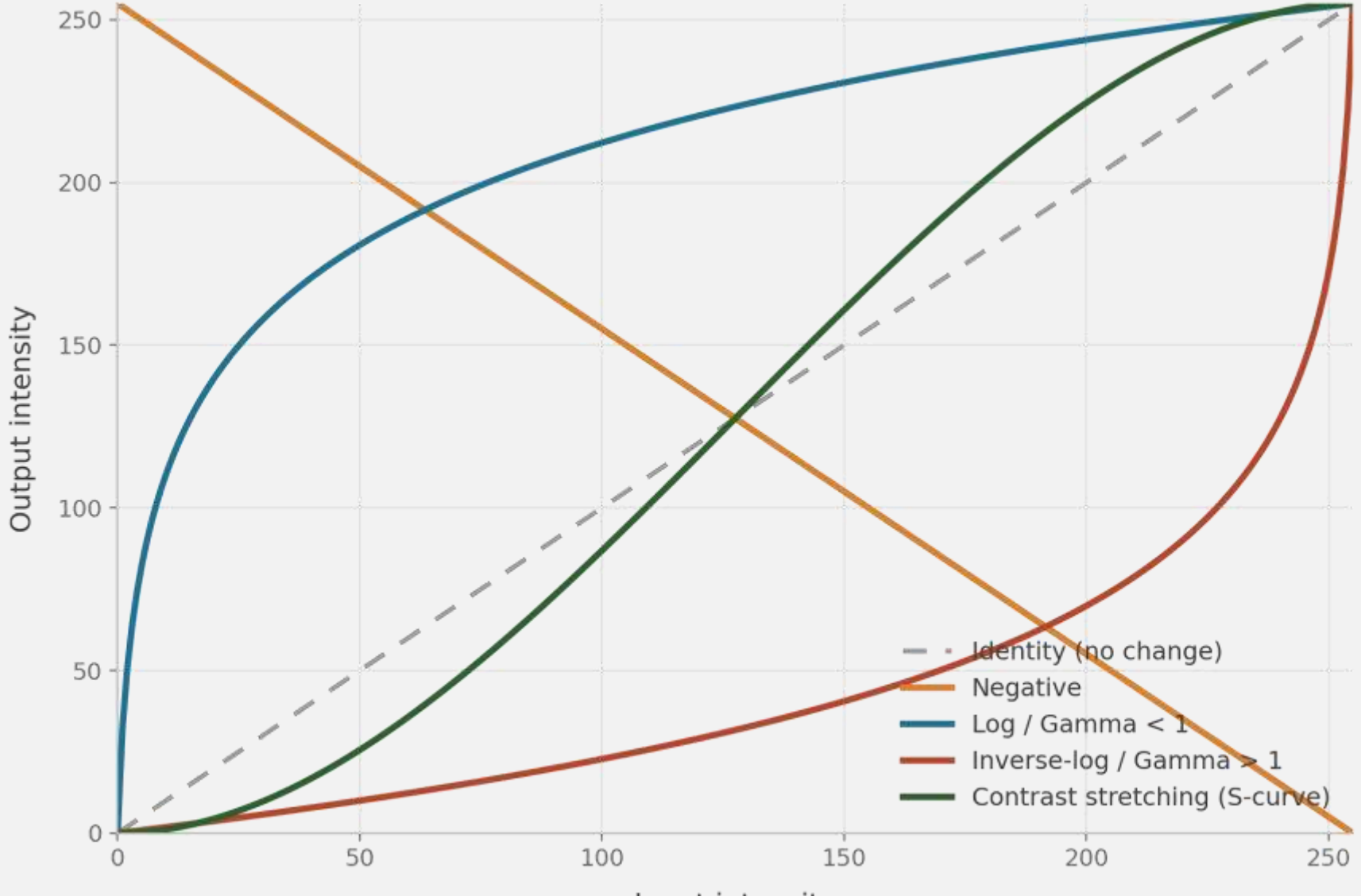
Analogy: rather than a flashlight or elastic band applying the same rule everywhere, imagine an editor who studies your photo first, then decides exactly how much to stretch each brightness range.

Best use: general-purpose auto-contrast enhancement — medical imaging and satellite photos, where the ideal range isn't known in advance.

The key difference: its curve is the image's own cumulative distribution function (CDF) — different for every image, unlike the fixed formulas on the other slides.

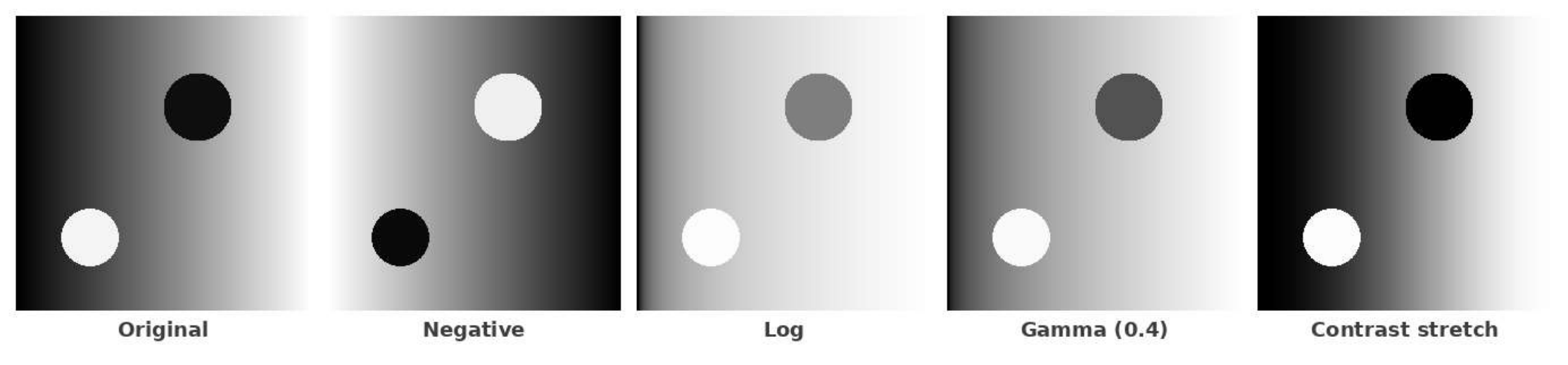
All five curves, side by side

The dashed gray line is the identity baseline — every other transform is a bend away from it.



Same image, five transforms

A synthetic test image with a dark circle, a bright circle, and a gradient between them.



Log and gamma both pull shadow detail up; contrast stretch pushes the midtones harder toward black or white; negative flips everything.

Quick reference

Transform	Effect	Best for
Negative	Flips all tones	Hidden detail in dark regions
Log	Boosts shadows only	High dynamic range data
Gamma	Tunable bright/dark	Display calibration
Contrast stretch	Expands midtone contrast	Washed-out photos
Histogram equalization	Auto-balances per image	Unknown ideal range

Point transformations are fixed formulas applied uniformly. Histogram equalization is the one exception — it's adaptive, shaped by the image itself.